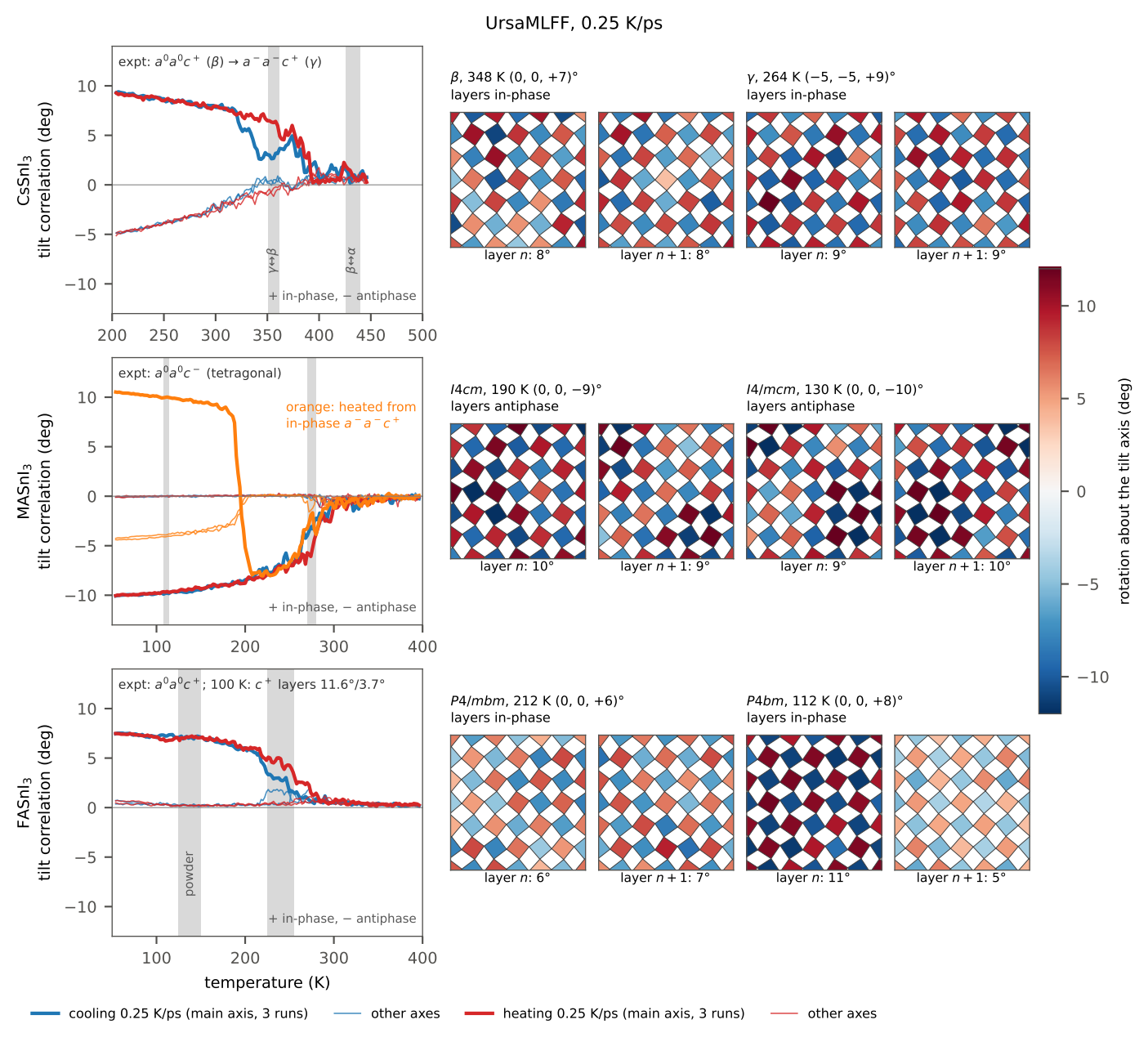

UrsaMLFF, 0.25 K/ps
expt: $a^0a^0c^+$ (β) → $a^-a^-c^+$ (γ)
CsSnI$_3$ tilt correlation (deg)
$\gamma\to\beta$
$\beta\to\alpha$
+ in-phase, − antiphase
β, 348 K (0, 0, +7)°
layers in-phase
γ, 264 K (−5, −5, +9)°
layers in-phase
layer n: 8°
layer $n+1$: 8°
layer n: 9°
layer $n+1$: 9°
expt: $a^0a^0c^-$ (tetragonal)
orange: heated from in-phase $a^-a^-c^+$
MASnI$_3$ tilt correlation (deg)
+ in-phase, − antiphase
$I4cm$, 190 K (0, 0, −9)°
layers antiphase
$I4/mcm$, 130 K (0, 0, −10)°
layers antiphase
layer n: 10°
layer $n+1$: 9°
layer n: 9°
layer $n+1$: 10°
expt: $a^0a^0c^+$; 100 K: c^+ layers 11.6°/3.7°
FASnI$_3$ tilt correlation (deg)
powder
+ in-phase, − antiphase
$P4/mbm$, 212 K (0, 0, +6)°
layers in-phase
$P4bm$, 112 K (0, 0, +8)°
layers in-phase
layer n: 6°
layer $n+1$: 7°
layer n: 11°
layer $n+1$: 5°
rotation about the tilt axis (deg)
temperature (K)
cooling 0.25 K/ps (main axis, 3 runs)
other axes
heating 0.25 K/ps (main axis, 3 runs)
other axes